

POOJA SUKHDEVE

Data Scientist | Machine Learning Engineer | AI & Predictive Analytics

poojasukhdeve06@gmail.com | Remote-Ready · Global Opportunities | [LinkedIn](#) | [Portfolio](#)

PROFESSIONAL SUMMARY

Data Scientist and Machine Learning Engineer with 6+ years of experience transforming complex, multi-source datasets into production-ready predictive models and actionable business intelligence. M.S. Computer Science (Data Analytics) from Boston University (GPA 3.59). Expert in Python, SQL, Scikit-learn, and Power BI, with hands-on delivery of end-to-end ML pipelines, SHAP-based explainability, NLP systems, and ETL architectures across healthcare, finance, and operations domains. Proven ability to evaluate competing models at scale (36+ models), communicate insights to non-technical stakeholders, and reduce processing time by up to 50%. Seeking Data Scientist and Applied AI roles to build interpretable, high-impact models that drive measurable decisions.

EDUCATION

M.S. Computer Science — Data Analytics Concentration | GPA: 3.59/4.00

May 2026

Boston University | Boston, MA

B.E. Computer Science Engineering

Jun 2016

Swami Ramanand Teerth Marathwada University | India

TECHNICAL SKILLS

Machine Learning & AI: Scikit-learn, Random Forest, XGBoost, PyTorch, TensorFlow, NLP, LLMs, GPT-4, Hugging Face, SHAP Explainability, Predictive Modelling, Feature Engineering, A/B Testing, OpenCV, OpenAI API

Data Analysis & Visualization: Python (Pandas, NumPy, Matplotlib, Seaborn), R, Power BI (DAX), Tableau, Data Wrangling, Statistical Analysis, Business Intelligence

Data Engineering & Cloud: ETL Pipelines, SQL, MySQL, PostgreSQL, MongoDB, Firebase, SQLite, Apache Spark, AWS, Docker, GitHub Actions, CI/CD

Programming & Software: Python, SQL, JavaScript, TypeScript, Flask, Node.js, React, Next.js, RESTful APIs, Git, Agile/Scrum

PROFESSIONAL EXPERIENCE

Data Research Analyst | *Boston University MET*

Sep 2024 – Present

- Designed and curated 1,000+ lines of interactive Python exercises using data-driven pedagogical methodologies, measurably improving course completion rates and learning accessibility for visually impaired learners.
- Applied data visualization best practices — diagrams, flowcharts, and tactile learning tools — to translate complex programming and data concepts into accessible, structured instructional content.
- Collaborated with faculty researchers to investigate inclusive education frameworks, authoring accessibility-compliant technical documentation in LaTeX and Overleaf aligned with WCAG standards.

Operations Analyst | *Boston University IS&T*

Sep 2024 – Oct 2025

- Engineered automated data workflows in Excel (VLOOKUP, pivot tables, VBA macros) across 1,000+ entry datasets, cutting reporting time by 40% and reducing data entry errors by 30%.
- Optimized end-to-end inventory and logistics tracking for 350+ shipments, achieving a 20% reduction in supply costs and zero stockouts through data-driven demand forecasting.
- Developed operational analytics reports enabling leadership to make evidence-based procurement and logistics decisions across the university's IS&T division.

Software Developer Intern | *ASVI Systems*

May 2025 – Aug 2025

- Architected and deployed 3 full-stack web applications (React.js, Next.js, Node.js) serving 500+ daily active users with secure JWT authentication and role-based access control.
- Engineered 8+ RESTful APIs processing 10,000+ daily requests via Express.js, Flask, MongoDB, and PostgreSQL — incorporating query optimization to maximize throughput and reliability.
- Achieved 35% page load time reduction through systematic performance profiling, code refactoring, and database query tuning.
- Automated CI/CD pipeline with Docker and GitHub Actions, compressing deployment time from 2 hours to 15 minutes and reducing production incidents by 60%.

Data Analyst | *Tech Zenesis*

Oct 2022 – Dec 2023

- Architected a scalable Python-based ETL pipeline (Pandas, NumPy, SQL) integrating 5+ heterogeneous data sources for 50+ business users, reducing data processing time by 50% and eliminating manual bottlenecks.

- Developed 8+ interactive Power BI dashboards with complex DAX calculations and real-time data refresh, surfacing KPIs across sales, marketing, and operations to senior stakeholders.
- Applied statistical analysis and predictive modelling techniques to identify revenue trends and operational inefficiencies, directly informing data-driven strategy and resource allocation decisions.
- Partnered cross-functionally with stakeholders to translate ambiguous business requirements into structured analytical frameworks and scalable reporting solutions.

Software Developer | *Interbeing Foundation*

Apr 2020 – Jun 2022

- Built and maintained 5+ full-stack applications (Python/Flask, Node.js/Express, JavaScript) supporting 10,000+ users and 50K+ monthly transactions at 99.9% uptime.
- Led Agile sprint cycles in collaboration with product managers and designers to ship 30+ features, boosting user engagement by 40% within six months.
- Improved application performance through systematic code refactoring and database query optimization, cutting average page load times by 45%.

Web Developer | *Oceans Software Technologies*

Apr 2018 – Mar 2020

- Designed and delivered responsive user interfaces for 8+ banking web applications (HTML5, CSS3, JavaScript) covering account management, loan applications, and transaction workflows.
- Implemented CSS media query-based layouts improving mobile user experience by 30% and achieving 95%+ cross-browser compatibility.
- Integrated frontend data-entry forms with MySQL and MongoDB backends via JavaScript validation and RESTful APIs, ensuring data integrity and smooth user flows.

DATA SCIENCE & AI PROJECTS

BRFSS Healthcare Predictive Analytics & Disease Risk Modeling | [GitHub](#)

- Constructed a production-grade healthcare ML platform on the CDC BRFSS dataset to predict chronic disease risk and support actuarial health insurance decisions — best-performing Random Forest achieved 84.3% classification accuracy.
- Systematically evaluated 36 predictive models applying rigorous statistical analysis, feature engineering, and hyperparameter tuning to identify optimal configurations for imbalanced healthcare data.
- Leveraged SHAP-based explainability to surface and rank key disease risk factors, translating complex model outputs into interpretable, clinician-ready risk narratives for non-technical stakeholders.

SPY ETF Machine Learning Trading Strategy | [GitHub](#)

- Engineered 10+ financial features (rolling returns, volatility metrics, moving averages, volume ratios) to feed three ML models — Linear Regression, Random Forest Regressor, and Random Forest Classifier — for next-day SPY ETF return prediction.
- Constructed a full backtesting framework measuring strategies against total return, Sharpe ratio, and equity curves; Random Forest Regressor delivered 8.60% cumulative return (Sharpe 0.737) vs. Buy & Hold benchmark at 24.99% (Sharpe 1.897).
- Applied rigorous model evaluation and realistic benchmarking methodology to translate quantitative results into clear, actionable insights on the practical limits and value of ML-driven trading strategies.

AI-Powered Restaurant Analytics Dashboard | [GitHub](#) | [Live Demo](#)

- Built an end-to-end AI analytics platform using Python, Flask, Chart.js, NLP, and the Hugging Face API to convert raw Zomato India restaurant data into actionable market intelligence.
- Designed interactive dashboards revealing ratings distributions, cuisine trends, and pricing patterns; integrated a conversational AI assistant delivering personalized recommendations via natural language queries.
- Applied data wrangling, feature engineering, and statistical analysis to surface customer preference signals and pricing dynamics across 10,000+ restaurant entries.

Chatalogue – University Chatbot Assistant | [GitHub](#)

- Designed and deployed an AI university assistant integrating GPT-4.1-mini, NLP, semantic search, and deterministic SQL generation to deliver accurate, context-aware responses for course schedules, instructor lookup, and campus navigation.
- Implemented a natural-language-to-SQL execution pipeline with multithreaded request handling and real-time data retrieval via OpenWeatherMap and OpenAI APIs, ensuring low-latency, reliable responses at scale.

PUBLICATIONS & ACHIEVEMENTS

- Winner - Boston University Hackathon 2024: Spearheaded development of a data science model that generated actionable insights and delivered measurable business impact, placing first among competing teams.
- Publication: Chapter 6 - Implementing Augmented Reality into Immersive Virtual Learning Environments, IGI Global, Pennsylvania, USA (2021).
- Publication: Chapter 11 - Handbook of Research on Driving Transformational Change in the Digital Built Environment, IGI Global, Pennsylvania, USA (2021).